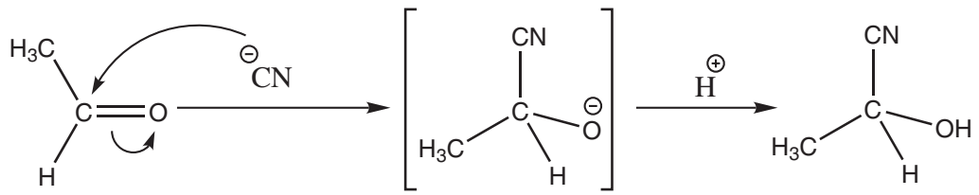
Examples

1) Addition of HCN

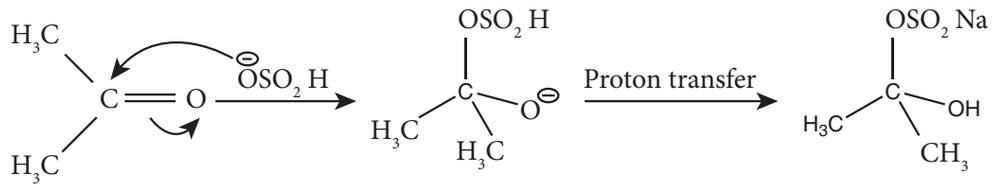
Attack of CN^- on carbonyl carbon followed by protonation gives cyanohydrins.



ethanal (acetaldehyde) | tetrahedral intermediate | 2-hydroxy propanenitrile (acetaldehyde cyano hydrin)

The cyanohydrins can be converted into hydroxy acid by acid hydrolysis. Reduction of cyanohydrins gives hydroxy amines

2) Addition of $NaHSO_3$



acetone (propanone) | bisulphite addition compound

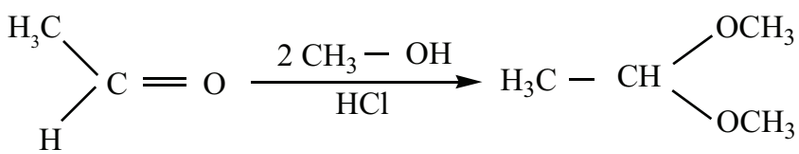
This reaction finds application in the separation and purification of carbonyl compound. The bisulphate addition compound is water soluble and the solution is treated with mineral acid to regenerate the carbonyl compounds .

3) Addition of alcohol

When aldehydes / ketones is treated with 2 equivalents of an alcohol in the presence of an acid catalyst to form acetals.

example

When acetaldehyde is treated with 2 equivalent of methanol in presence of HCl, 1,1, - dimethoxy ethane is obtained.



ethanal (acetaldehyde) | 1,1 - dimethoxyethane (acetaldehyde dimethyl acetal)

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